REVIEW OF EIGHT STANDARD PROPERTIES OF AN ABELIAN VARIETY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file groupoids.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Editorial note. Occupation scan: this chapter's pool has 47 of 128 labels already delivered by the S7 line (S7-gp-* registrations, statement_anchor-prose scan; withdrawn cards excluded); none of the 11 accepted labels of this batch is among them (gp-occupied.json, 2026-10-14).

Source reference. Original lemma 0BFA: sources/stacks-chow/tags.txt, source lines 14378–14378. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0BFD: sources/stacks-chow/tags.txt, source lines 14381–14381. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0BFG: sources/stacks-chow/tags.txt, source lines 14384–14384. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0BFH: sources/stacks-chow/tags.txt, source lines 14385–14385. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 02GV: sources/stacks-chow/tags.txt, source lines 3242–3242. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0475: sources/stacks-chow/tags.txt, source lines 5496–5496. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0C0Y: sources/stacks-chow/tags.txt, source lines 15103–15103. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Proposition 0.1. *Let A be an abelian variety over a field k. Then*

(1) *A is projective over k,*
(2) *A is a commutative group scheme,*
(3) *the morphism $[n] : A \to A$ is surjective for all $n \geq 1$,*
(4) *if k is algebraically closed, then $A(k)$ is a divisible abelian group,*

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(5) $A[n] = \mathrm{Ker}([n] : A \to A)$ *is a finite group scheme of degree* $n^{2\dim A}$ *over* k,
(6) $A[n]$ *is étale over* k *if and only if* $n \in k^*$,
(7) *if* $n \in k^*$ *and* k *is algebraically closed, then* $A(k)[n] \cong (\mathbf{Z}/n\mathbf{Z})^{\oplus 2\dim(A)}$,
(8) *if* k *is algebraically closed of characteristic* $p > 0$, *then there exists an integer* $0 \leq f \leq \dim(A)$ *such that* $A(k)[p^m] \cong (\mathbf{Z}/p^m\mathbf{Z})^{\oplus f}$ *for all* $m \geq 1$.

Proof. Part (1) follows from Lemma 0BFA. Part (2) follows from Lemma 0BFD. Part (3) follows from Lemma 0BFG. If k is algebraically closed then surjective morphisms of varieties over k induce surjective maps on k-rational points, hence (4) follows from (3). Part (5) follows from Lemma 0BFG and the fact that a base change of a finite locally free morphism of degree N is a finite locally free morphism of degree N. Part (6) follows from Lemma 0BFH. Namely, if n is invertible in k, then $[n]$ is étale and hence $A[n]$ is étale over k. On the other hand, if n is not invertible in k, then $[n]$ is not étale at e and it follows that $A[n]$ is not étale over k at e (use Morphisms, Lemmas 02GV and 0475).

Assume k is algebraically closed. Set $g = \dim(A)$. Proof of (7). Let ℓ be a prime number which is invertible in k. Then we see that

$$A[\ell](k) = A(k)[\ell]$$

is a finite abelian group, annihilated by ℓ, of order ℓ^{2g}. It follows that it is isomorphic to $(\mathbf{Z}/\ell\mathbf{Z})^{2g}$ by the structure theory for finite abelian groups. Next, we consider the short exact sequence

$$0 \to A(k)[\ell] \to A(k)[\ell^2] \xrightarrow{\ell} A(k)[\ell] \to 0$$

Arguing similarly as above we conclude that $A(k)[\ell^2] \cong (\mathbf{Z}/\ell^2\mathbf{Z})^{2g}$. By induction on the exponent we find that $A(k)[\ell^m] \cong (\mathbf{Z}/\ell^m\mathbf{Z})^{2g}$. For composite integers n prime to the characteristic of k we take primary parts and we find the correct shape of the n-torsion in $A(k)$. The proof of (8) proceeds in exactly the same way, using that Lemma 0C0Y gives $A(k)[p] \cong (\mathbf{Z}/p\mathbf{Z})^{\oplus f}$ for some $0 \leq f \leq g$. □